

The easyMelts geobarometer

User guide for phase-equilibrium geobarometry with rhyolite-MELTS

easyMelts 0.3.0 with the Geobarometer tab, release v0.3.0-geobarometer.3, Windows and macOS, September 2026

1. What it does

The **Geobarometer** tab finds the pressure at which a melt composition, usually a glass analysis, is saturated at its liquidus in the minerals you choose: quartz and two feldspars, or plagioclase, quartz and orthopyroxene, for example. It is the phase-equilibrium geobarometer of Gualda & Ghiorso (2014) and Harmon et al. (2018), calculated the way the *Qtz + Fspars P Calc* of MELTS_Excel does it, with the same rhyolite-MELTS 1.0.x model. Everything runs on your computer: no Excel, no internet connection and no web service.

On top of what the spreadsheet does, it runs several fO_2 values in one go, reads batches of compositions from a CSV file, plots the saturation curves and residuals, exports the results, and comes with a command-line version for large batches.

2. Installing and starting

Use release v0.3.0-geobarometer.3 or later. In v0.3.0-geobarometer.1 the easyMelts window ran rhyolite-MELTS without its adjustment to sanidine and gave wrong saturation temperatures: for the Bishop Tuff sample it found no pressure instead of 377.3 and 372.5 MPa (the command-line tool of that release was right). Up to v0.3.0-geobarometer.2, MELTS could crash part way through a batch on glasses that contain none of an oxide, MgO- or TiO₂-free glasses for example (section 11). Release .3 also reads conditions for each row of a batch file, and runs large batches in parallel (section 6).

- Unzip `easyMelts_geobarometer_win64.zip` into any folder, for example `Documents\easyMelts`, and keep the files together.
- Double-click `easyMelts.exe`. Windows may say the program comes from an unknown publisher: click *More info*, then *Run anyway*.
- A black console window opens with the program. It shows MELTS messages and can be ignored; closing it closes easyMelts.
- Exported files are written to the folder that contains `easyMelts.exe`.
- On a Mac, unzip the `arm64` (Apple silicon) or `x86_64` (Intel) zip and follow `README-macOS.txt`; the program then works as described here.

File	What it is
<code>easyMelts.exe</code>	easyMelts with the Geobarometer tab
<code>geobarometer_cli.exe</code>	the same calculation from the command line (section 10)
<code>glasses_template.csv</code>	example of a batch file (section 6)
<code>conditions_template.csv</code>	example of a batch file with conditions for each row (section 6)
<code>run_batch_example.bat</code>	runs a batch file in parallel with the command-line tool (section 6)
<code>GEOBAROMETER.md</code>	short technical notes
<code>LICENSE.txt</code> , <code>source/geobarometer.patch</code>	licence and the source changes (section 12)

3. A first calculation

1. **Init > Version > MELTS_v1.0.x.** The published geobarometer uses rhyolite-MELTS 1.0.x; the tab flags any other version in orange.
2. In the **Input/Output** tab open *Set initial composition* and type the glass composition as analysed (wt%). Put **H₂O = 13**: the excess water keeps the melt saturated with fluid at every pressure of the grid, as in Ruefer et al. (2025). The FeO/Fe₂O₃ split does not matter, because the fO₂ buffer resets it; enter FeO_t as FeO. To try the program first, use *Samples > Bishop Tuff*, the test composition of MELTS_Excel.
3. Open the **Geobarometer** tab and keep *Input/Output tab* as the source.
4. Check the settings (section 4). The defaults are those of MELTS_Excel.
5. Click **Run geobarometer**. A window counts the pressure paths; *Stop* ends the run early and keeps what has finished.
6. The results table appears on the right. Click a row to see its plots (Figure 1).

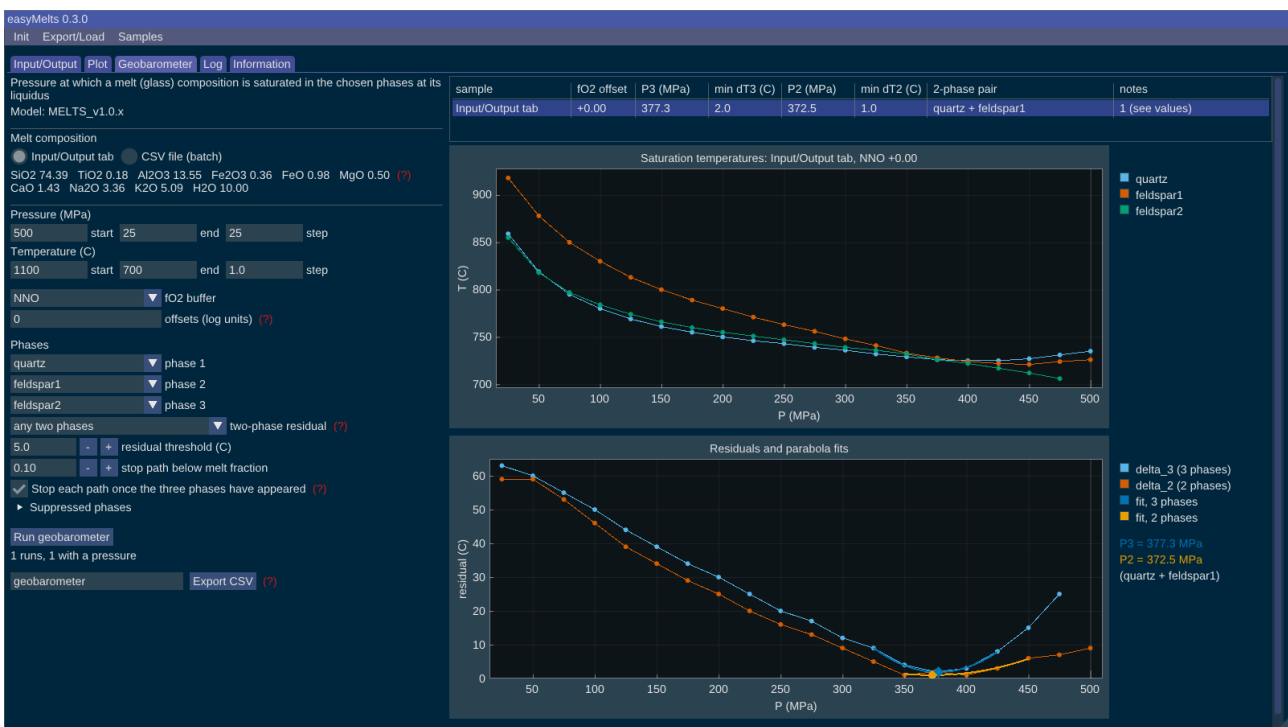


Figure 1. The Bishop Tuff sample with the default settings (about 40 s): three-phase pressure P3 = 377.3 MPa, two-phase pressure P2 = 372.5 MPa with quartz and feldspar1 first to crystallise. Top: saturation temperature of each phase against pressure. Bottom: the residuals, the fitted parabolas and the resulting pressures.

4. Settings

Pressure at which a melt (glass) composition is saturated in the chosen phases at its liquidus
Model: MELTS_v1.0.x

Melt composition
☒ Input/Output tab ☐ CSV file (batch)
SiO₂ 74.39 TiO₂ 0.18 Al₂O₃ 13.55 Fe₂O₃ 0.36 FeO 0.98 MgO 0.50 (?)
CaO 1.43 Na₂O 3.36 K₂O 5.09 H₂O 10.00

Pressure (MPa)
500 start 25 end 25 step

Temperature (°C)
1100 start 700 end 1.0 step

NNO fO₂ buffer
-1, 0, 0.5 offsets (log units) (?)

Phases
quartz phase 1
feldspar phase 2
orthopyroxene phase 3
any two phases two-phase residual (?)

5.0 - + residual threshold (°C)
0.10 - + stop path below melt fraction

☒ Stop each path once the three phases have appeared (?)
▶ Suppressed phases

Run geobarometer
3 runs, 2 with a pressure
geobarometer Export CSV (?)

Figure 2. The settings panel with quartz, feldspar and orthopyroxene and three NNO offsets.

Setting	Default	What it does
Melt composition	Input/Output tab	Use the composition of the Input/Output tab, or a CSV file of many compositions (section 6).
Pressure start, end, step	500, 25, 25 MPa	The pressure grid, calculated from the higher to the lower value. Ruefer et al. (2025) reran compositions that gave pressures below 50 MPa with 10 MPa steps.
Temperature start, end, step	1100, 700, 1 °C	The temperature grid of the cooling paths. Saturation temperatures are resolved to this step.
fO ₂ buffer	NNO	None, HM, NNO, QFM, COH or IW.
offsets (log units)	0	One run per value, separated by commas, for example -1, -0.75, -0.5, 0, 0.5.
phase 1, 2, 3	quartz, feldspar1, feldspar2	The three minerals whose saturation curves are compared (names in section 5).
two-phase residual	any two phases	How the two-phase residual is formed (section 8).
residual threshold (°C)	5	A pressure is reported only if the smallest residual is at or below this value. Part 5 of the series used 10 °C for magnetite-bearing assemblages.
stop path below melt fraction	0.10	A cooling path ends when melt is less than this mass fraction of the system, as in MELTS_Excel.
Stop each path once the three phases have appeared	on	Saves time and cannot change any result (section 8). Untick it to follow every path to the end.
Count negative oxide values as 0	on	A negative amount of an oxide, which MELTS cannot use, counts as 0 and the notes say so. Untick it to leave such compositions out.
Suppressed phases	amphibole, biotite	Phases MELTS may not crystallise (the MELTS_Excel default). The list in the Input/Output tab does not apply here.

Reproducing a MELTS_Excel run

The `init_cond` sheet of a MELTS_Excel workbook holds everything the tab needs. Enter it as follows, and the tab gives the saturation temperatures of the workbook's Phase_Data sheet and the pressures of its P_Calc sheet.

MELTS_Excel	Geobarometer tab
Model rhyolite-MELTS_v1.0.x	Init > Version > MELTS_v1.0.x
composition, including H ₂ O	Input/Output tab (Set initial composition), or one row of a CSV file
T1, T2, T step (°C)	Temperature start, end, step
P1, P2, P step (MPa)	Pressure start, end, step
fO ₂ buffer, fO ₂ value	fO ₂ buffer, offsets
P_Calc sheet: quartz, feldspar1, feldspar2	the default phases, with <i>any two phases</i>

If the pressures were worked out from the saturation temperatures with other phases or with the *require phase 1* rule, for example feldspar1, quartz and orthopyroxene, enter the phases in the same order. Under *require phase 1* the two-phase residual always involves phase 1, so the order changes the two-phase pressure (the three-phase pressure does not depend on it).

5. Reading the results

Results table

Column	Meaning
sample, fO ₂ offset	The composition (the run title, or the name in the CSV file) and the fO ₂ offset of this run.
P3 (MPa)	Pressure at which the three phases saturate together; '-' when none is found.
min dT3 (°C)	Smallest three-phase residual on the grid. Above the threshold, no three-phase pressure is reported.
P2 (MPa), min dT2 (°C)	The same for the two-phase residual.
2-phase pair	The two phases that set the two-phase residual at its minimum, i.e. the pair the pressure P2 refers to.
notes	Number of pressures with a warning. Open <i>Values at each pressure</i> below the plots to read them.

Plots

The upper plot shows the saturation temperature of each chosen phase against pressure; the lower plot shows the two residuals, the parabolas fitted near their minima, and a diamond at each pressure found. The colour key, the pressures and, while the mouse is over a plot, the cursor position are listed to the right of each plot, so no text covers the data. Drag to pan, scroll to zoom, and double-click to fit the data again.



Figure 3. Results of the run of Figure 2 with the NNO +0 row selected: plagioclase and orthopyroxene are the first phases to crystallise near 60 MPa, and no three-phase pressure is found (smallest residual 58 °C).

Phase names

Name	Meaning
feldspar	Any feldspar.
feldspar1, feldspar2	The first feldspar to appear and the second coexisting feldspar, numbered as in MELTS_Excel.
plagioclase, sanidine	Feldspar with less than, or at least, 25 mol% sanidine component (the MELTS_Excel split).
quartz, orthopyroxene, clinopyroxene, spinel, rhm-oxide, ...	Any MELTS phase. Solution phases can also be chosen as, for example, clinopyroxene1 and clinopyroxene2.

Notes

Note	Meaning
wet liquidus not found, path starts at T start	The fluid-saturated liquidus search failed at that pressure, so the path starts at the top of the temperature grid (MELTS_Excel does the same). Results at that pressure are normally unaffected.
MELTS did not converge within 20 s at T °C, path stopped	One equilibrium calculation did not converge. The path ends there, and phases that would have appeared further down are missing at that pressure.
MELTS failed at T °C, path stopped	MELTS returned an error; same consequence.
<phase> already present at the first step	The phase was already present at the first temperature of the path, so its true saturation temperature may be higher than recorded.

Note	Meaning
2-phase vertex outside the fitted points (summary file)	The minimum of the parabola lies outside the pressures used for the fit; treat that pressure with caution.
negative <oxide> (value) set to 0	The row had a negative amount of an oxide (a Monte Carlo draw below zero, for example), which MELTS cannot use, so it counted as 0. Untick <i>Count negative oxide values as 0</i> (negative=skip) to leave such rows out.
no FeO or Fe2O3: an fO2 buffer cannot be applied, not calculated	An fO2 buffer acts through the iron oxides; choose buffer <i>none</i> for an iron-free melt.
no result at any pressure (first, P MPa: ...)	No pressure gave a result; the first failure is quoted (for example, the path starts below T end when the wet liquidus is lower than the end of the temperature grid).

6. Many compositions (CSV batch)

Prepare a table with one row per composition and a header row of oxide names, and save it from Excel as *CSV UTF-8* or *CSV (comma delimited)*. The template that comes with the program looks like this:

```
Sample,SiO2,TiO2,Al2O3,Fe2O3,FeO,MnO,MgO,CaO,Na2O,K2O,P2O5,H2O
Bishop Tuff (MELTS_Excel default),74.39,0.18,13.55,0.36,0.976,0,0.5,1.43,3.36,5.09,0,10
```

- Recognised columns: SiO₂, TiO₂, Al₂O₃, Fe₂O₃, Cr₂O₃, FeO, MnO, MgO, NiO, CoO, CaO, Na₂O, K₂O, P₂O₅, H₂O, CO₂. FeOt, FeO* and FeOT are read as FeO. The first other column holds the sample name.
- Commas, semicolons (with decimal commas) and tabs all work; empty cells count as zero.
- Rows without H₂O get the amount in *H2O added when missing* (13 g by default) added to the analysis.

Then put the file next to easyMelts.exe, choose *CSV file (batch)*, pick the file in *CSV in this folder* (or type its full path), click **Load** and **Run geobarometer**. Each composition runs at every fO₂ offset, so the table has one row per composition and offset. A run of 20 pressures takes about 40 s on a laptop, so 53 runs take about 40 minutes.

Conditions for each row

A row can also carry its own conditions, so that one file holds runs at different pressures, temperatures, fO₂ and phase assemblages, as the multiple-compositions sheet of MELTS_Excel does. Add any of the columns below; a blank cell takes the value set in the tab.

Column	Also read as	Meaning
P_start, P_end, P_step	P1 (MPa), P2 (MPa), P step (MPa)	pressure grid (MPa)
T_start, T_end, T_step	T1 (C), T2 (C), T step (C)	temperature grid (°C)
buffer	fO2 buffer	none, HM, NNO, QFM, COH or IW
offsets	fO2 value, offset	one or more fO ₂ offsets, separated by spaces, or ;, for example -1 -0.5 0
phase1, phase2, phase3	Phase 1, Phase 2, Phase 3	the three phases, in the order that matters for <i>require phase 1</i>
rule	Formula	any (any two phases) or phase1 (require phase 1)
threshold	residual threshold (C)	residual threshold (°C)

conditions_template.csv has two Bishop Tuff rows: one with the settings of the tab, and one with its own grid, three offsets, feldspar1, quartz and orthopyroxene and *require phase 1*. The Load message says how many rows carry their own conditions; a condition that cannot be read (an unknown phase or buffer, or text where a number should be) is reported with its row, and nothing is loaded.

Large batches and ensembles

The window runs a batch one composition after another, and writes the summary of every finished composition to `geobarometer_autosave_summary.csv` as it goes, so that a long batch is not lost if the program is stopped. For thousands of compositions, a Monte Carlo ensemble for example, the command-line tool with `jobs=N` is much faster: it cuts the file into N parts, runs them at once, one per processor core, and joins the results into one summary and one detail file, in the order of the input file. Open a command prompt in the program folder and type, for example:

```
geobarometer_cli.exe ensemble.csv jobs=8 p_start=600 p_end=25 t_end=650 rule=phase1 out=ensemble
```

`run_batch_example.bat` does the same with a double-click: set the file name and the settings at its top in Notepad. It uses one part per processor core, keeping one core free. The settings are the keys of section 10, and the condition columns above still apply row by row.

7. Exporting results

Type a name in the box next to **Export CSV** and click it. Two files are written next to `easyMelts.exe` and open directly in Excel:

File	Contents
<code>name_summary.csv</code>	One row per composition and offset: <code>P_3phase_MPa</code> , <code>min_dT_3phase_C</code> , <code>P_at_min_3phase_MPa</code> , <code>points_3phase</code> , <code>P_2phase_MPa</code> , <code>min_dT_2phase_C</code> , <code>P_at_min_2phase_MPa</code> , <code>points_2phase</code> , <code>phases_2phase</code> , the three phases, the two-phase rule, the threshold, the number of equilibrium calculations, the notes, and for each parabola the pressures it was fitted to (<code>fit_P_min</code> , <code>fit_P_max</code>) and its coefficients (<code>fit_a</code> , <code>fit_b</code> , <code>fit_c</code> : residual in °C = $a P^2 + b P + c$, P in MPa).
<code>name_detail.csv</code>	One row per composition, offset and pressure: wet liquidus, the saturation temperature of the three phases, <code>delta_3</code> , <code>delta_2</code> , the parabolas at the pressures they were fitted to (<code>fit_3</code> , <code>fit_2</code>), the saturation temperature of every other phase that appeared (<code>Tsat_</code> columns) and the notes. With the summary, every curve of the tab can be redrawn.

8. How the pressure is calculated

1. For each pressure of the grid, MELTS finds the fluid-saturated liquidus of the composition.
2. The melt is then cooled in steps of dT , one equilibrium calculation per step, from the first grid temperature at or above that liquidus, $T_{\text{start}} - \text{Int}((T_{\text{start}} - T_{\text{liquidus}})/dT) dT$. Every pressure starts from a fresh system with the analysed composition. A path stops below the melt-fraction limit, at the end of the temperature grid, or if MELTS fails.
3. The saturation temperature of a phase is the highest temperature of the path at which the phase is present.
4. From the saturation temperatures T_1 , T_2 , T_3 of the three phases, two residuals are formed at each pressure: $\Delta T_3 = \max - \min$ of the three (only when all three are present), and $\Delta T_2 = \text{highest} - \text{second highest}$ (*any two phases*) or $|T_1 - \max(T_2, T_3)|$ (*require phase 1*). An absent phase counts as 0, as in the spreadsheet.
5. A parabola $\Delta T = aP^2 + bP + c$ is fitted by least squares through the smallest residual of the grid and up to two grid points on each side. The pressure is its vertex, $P = -b/(2a)$, reported only if the smallest residual is at or below the threshold and the parabola opens upwards ($a > 0$).

With *any two phases*, a two-phase pressure appears only where those two phases are the first to crystallise. A crossing of two saturation curves below a third phase is ignored, because the melt at that crossing would no longer have the analysed composition (Ruefer et al. 2025; Smithies et al. 2025). When a phase is absent from the natural assemblage, as quartz is at Cordón Caulle, pressures that involve it are maxima. Quartz and feldspar saturation hardly depend on fO_2 , while orthopyroxene and the oxides do, which is why several offsets are run.

Because only the highest temperature at which each phase appears is used, the rest of a path cannot change the result; *Stop each path once the three phases have appeared* therefore only saves time.

9. Checks against MELTS_Excel

The requests that MELTS_Excel (version of 11 August 2025) sends to its MELTS web service were replayed for its Bishop Tuff test composition at NNO, at 12 pressures, and compared with this program. Of 116 phase saturation temperatures, 115 are identical; the exception is the fluid at 25 MPa, 1 °C apart, because the two paths start one grid step apart there. The wet liquidus agrees to 0.01 °C at every pressure but 25 MPa, where the two searches stop 0.4 °C apart. That search repeats an equilibration and a liquidus calculation until two estimates agree within 0.5 °C, so its result depends on its route: from the 1100 °C start the two programs take different routes, and from a 930 °C start both return 918.87 °C. It only sets where a cooling path starts. At 475 MPa both fail to find the wet liquidus and start the path at 1100 °C. The Windows, Linux and macOS builds give identical tables, and the Geobarometer tab gives the same tables as the command-line tool: runs replayed through the tab on Windows match it value for value, and a test of the tab compiled exactly like the released program checks the Bishop Tuff result on every change.

P (MPa)	wet liquidus, Excel (°C)	wet liquidus, easyMelts (°C)	quartz	feldspar1	feldspar2
500	842.97	842.97	735	726	-
475	not found	not found	731	724	706
450	838.48	838.48	727	721	712
425	838.67	838.67	725	722	717
400	838.87	838.87	725	724	722
375	839.26	839.26	726	728	726
350	839.65	839.65	729	733	732
325	839.84	839.84	732	741	736
300	840.23	840.23	736	748	739
200	843.55	843.55	750	780	755
100	854.30	854.30	780	830	784
25	918.75	919.14	859	918	855

Table 1. Bishop Tuff, NNO. Saturation temperatures (°C) of quartz and both feldspars are the same in MELTS_Excel's web service and in easyMelts at every pressure compared.

Speed

One composition at one fO_2 (20 pressures, 500 to 25 MPa) takes 43 s on a laptop under Windows. MELTS_Excel sends the same work to its web service as 2,328 requests (a wet-liquidus search per pressure and one request per cooling step); replaying one complete MELTS_Excel run for the Bishop Tuff composition took 33 minutes of server time (0.85 s per request, measured on 26 September 2026), before any time Excel spends writing its sheets. The 53 runs of Ruefer et al. (2025) take about 40 minutes here, against at least 29 hours of web-service time.

Differences from MELTS_Excel

- An equilibrium calculation that has not converged after 20 s stops that pressure path, with a note. MELTS_Excel gives up on a web request after 2 s and a 20 s retry. Converged calculations take well under a second.
- A chosen phase that never appears is reported as absent at those pressures; the P_Calc sheet of MELTS_Excel shows errors or drops the residual in that case.
- Versions of MELTS_Excel differ in the two-phase residual: older ones compared quartz and feldspar1 only, the version of August 2025 uses the two highest saturation temperatures, and that of November 2025 adds *require phase 1*. This program offers the two current rules.
- MELTS no longer writes `melts.out` and a `tables` folder on every step.

10. Command-line version

geobarometer_cli.exe runs the same calculation without the window, which suits large batches. Open a command prompt in the program folder and type, for example:

```
geobarometer_cli.exe offsets=-1,-0.75,-0.5,0,0.5 phases=quartz,feldspar,orthopyroxene glasses.csv
geobarometer_cli.exe --bishop out=test
```

Key	Default	Meaning
p_start, p_end, p_step	500, 25, 25	pressure grid (MPa)
t_start, t_end, t_step	1100, 700, 1	temperature grid (°C)
buffer, offsets	nno, 0	fO ₂ buffer (none, hm, nno, qfm, coh, iw) and offsets
phases, rule	quartz,feldspar1,feldspar2, any	the three phases, and <i>any</i> or <i>phase1</i> for the two-phase residual
threshold, h2o	5, 13	residual threshold (°C); H ₂ O added to rows without it (g)
suppress, stop	amphibole,biotite, 1	suppressed phases; 1 stops a path once the three phases appear
step_timeout, version	20, 1.0.2	seconds allowed per equilibrium calculation; MELTS calibration (1.0.2, 1.1.0, 1.2.0)
out	geobarometer	prefix of the two output files
jobs	1	run the file in this many parts at once (one per processor core) and join the results (section 6)
negative	zero	a negative oxide amount counts as 0 (zero), or the composition is not calculated (skip)

The condition columns of section 6 work here too.

11. Troubleshooting

What you see	What to do
'Initialise MELTS first'	Choose Init > Version > MELTS_v1.0.x.
P3 or P2 is '-'	The smallest residual (min dT columns) is above the threshold: no satisfactory pressure for that assemblage at that fO ₂ . Try the other offsets or phase choices rather than raising the threshold.
A run takes long	Each pressure path takes one to a few seconds. Use fewer pressures, or keep <i>Stop each path</i> on.
Results differ from an old Excel run	Check the MELTS_Excel version (two-phase rule), the grid, H ₂ O, the fO ₂ buffer and the suppressed phases. Web-service timeouts in Excel can also drop points.
easyMelts closes or stops responding part way through a batch	Update to v0.3.0-geobarometer.3, which fixes MELTS faults found with glasses that contain none of an oxide; the finished compositions are in geobarometer_autosave_summary.csv. If it still happens, run the same file with the command-line tool, which reports the composition it was on, and send that row and the settings.
Where are the exported files?	Next to easyMelts.exe.
Phases suppressed in the Input/Output tab still crystallise	The Geobarometer tab has its own list, under <i>Suppressed phases</i> .

12. Licence and changes

easyMelts is free software under the GNU General Public License, version 3 (LICENSE.txt). MELTS is copyright Mark S. Ghiorso and distributed under a BSD licence. This build adds the Geobarometer tab and the command-line program, and fixes faults in the MELTS library that could close easyMelts during a run or leave it running for ever. All changes are in source/geobarometer.patch and described in GEOBAROMETER.md. Source code and updates:

github.com/ebreard/easyMelts-geobarometer. easyMelts and this build are not associated with or endorsed by the authors of MELTS or MELTS_Excel. If results from this build are published, the easyMelts licence asks that the changes be described or the modified source made available.

References

- Gualda, G.A.R., Ghiorso, M.S., Lemons, R.V., Carley, T.L. (2012). Rhyolite-MELTS: a modified calibration of MELTS optimized for silica-rich, fluid-bearing magmatic systems. *Journal of Petrology* 53, 875-890.
- Gualda, G.A.R., Ghiorso, M.S. (2014). Phase-equilibrium geobarometers for silicic rocks based on rhyolite-MELTS. Part 1: Principles, procedures, and evaluation of the method. *Contributions to Mineralogy and Petrology* 168, 1033.
- Gualda, G.A.R., Ghiorso, M.S. (2015). MELTS_Excel: A Microsoft Excel-based MELTS interface for research and teaching of magma properties and evolution. *Geochemistry, Geophysics, Geosystems* 16, 315-324.
- Harmon, L.J., Cowlyn, J., Gualda, G.A.R., Ghiorso, M.S. (2018). Phase-equilibrium geobarometers for silicic rocks based on rhyolite-MELTS. Part 4: Plagioclase, orthopyroxene, clinopyroxene, glass geobarometer, and application to Mt. Ruapehu, New Zealand. *Contributions to Mineralogy and Petrology* 173, 7.
- Ruefer, A.C., Kelly, L.J., Gualda, G.A.R., Carrillo, E.L., Hickernell, S., Ward, S., Winslow, H., Ruprecht, P. (2025). In one step: Insights into shallow differentiation from basalt to rhyolite at Cordón Caulle from rhyolite-MELTS simulations. *Journal of Volcanology and Geothermal Research* 462, 108305.
- Smithies, S.L., et al. (2025). Phase-equilibrium geobarometers for silicic rocks based on rhyolite-MELTS. Part 5: Principles for multiple-phase geobarometry with examples from plagioclase + orthopyroxene $\pm$ quartz $\pm$ magnetite assemblages. *Contributions to Mineralogy and Petrology*, doi:10.1007/s00410-025-02274-w.